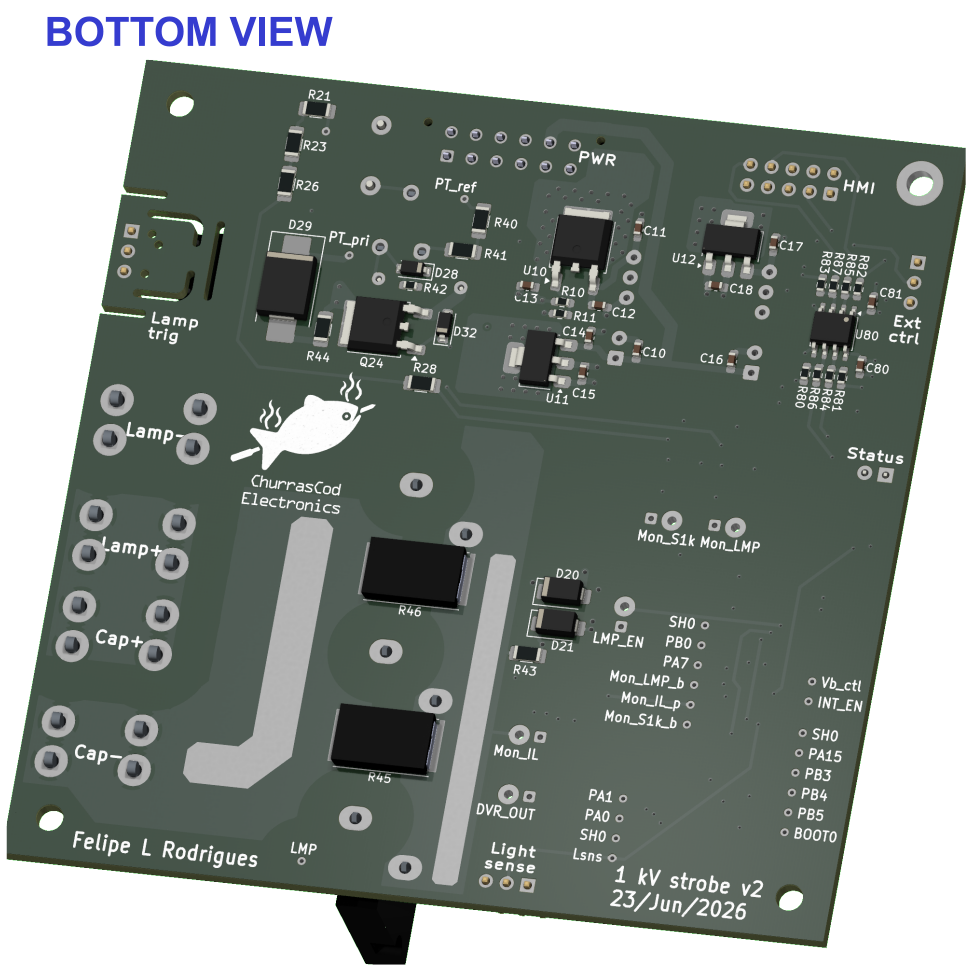
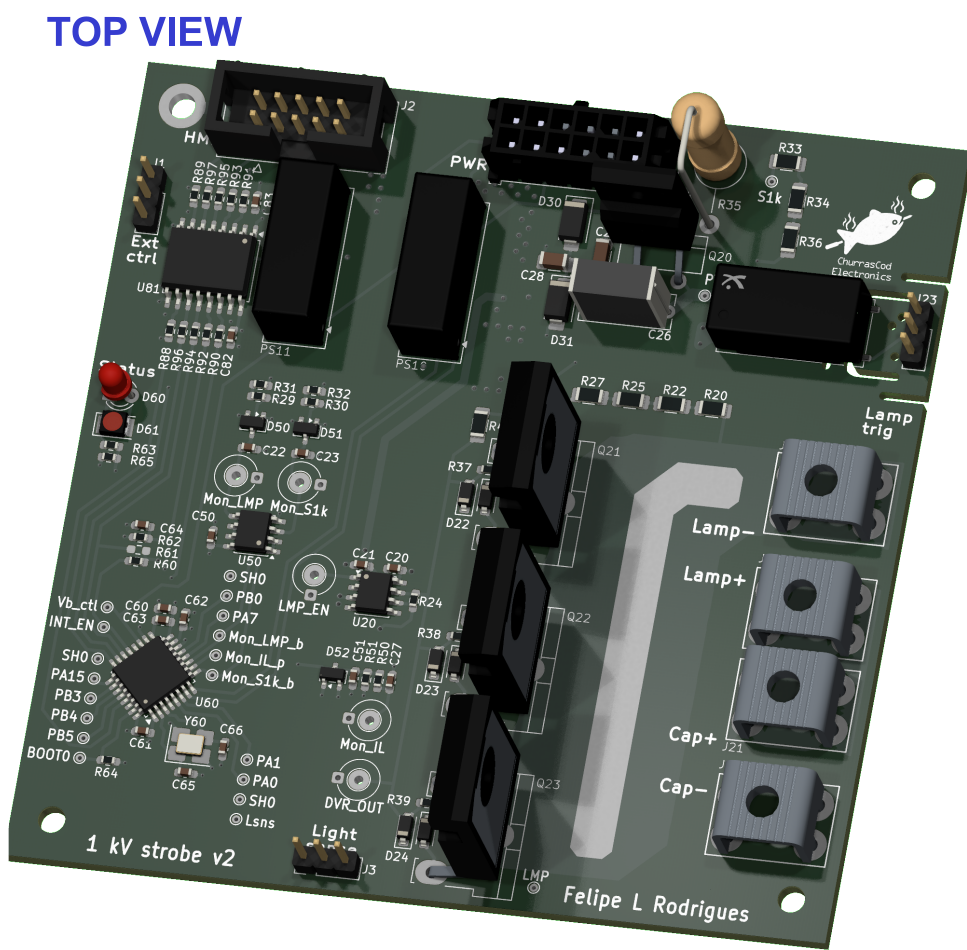


Main board

Rev 2.0.0+ (Unreleased)

2026-09-05

Page	Index
1	Cover Page
2	Block Diagram
3	Power supply
4	Lamp driver
5	Signal conditioning
6	MCU
7	Optical isolation
8	Revision History

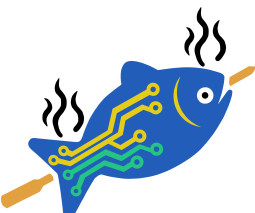


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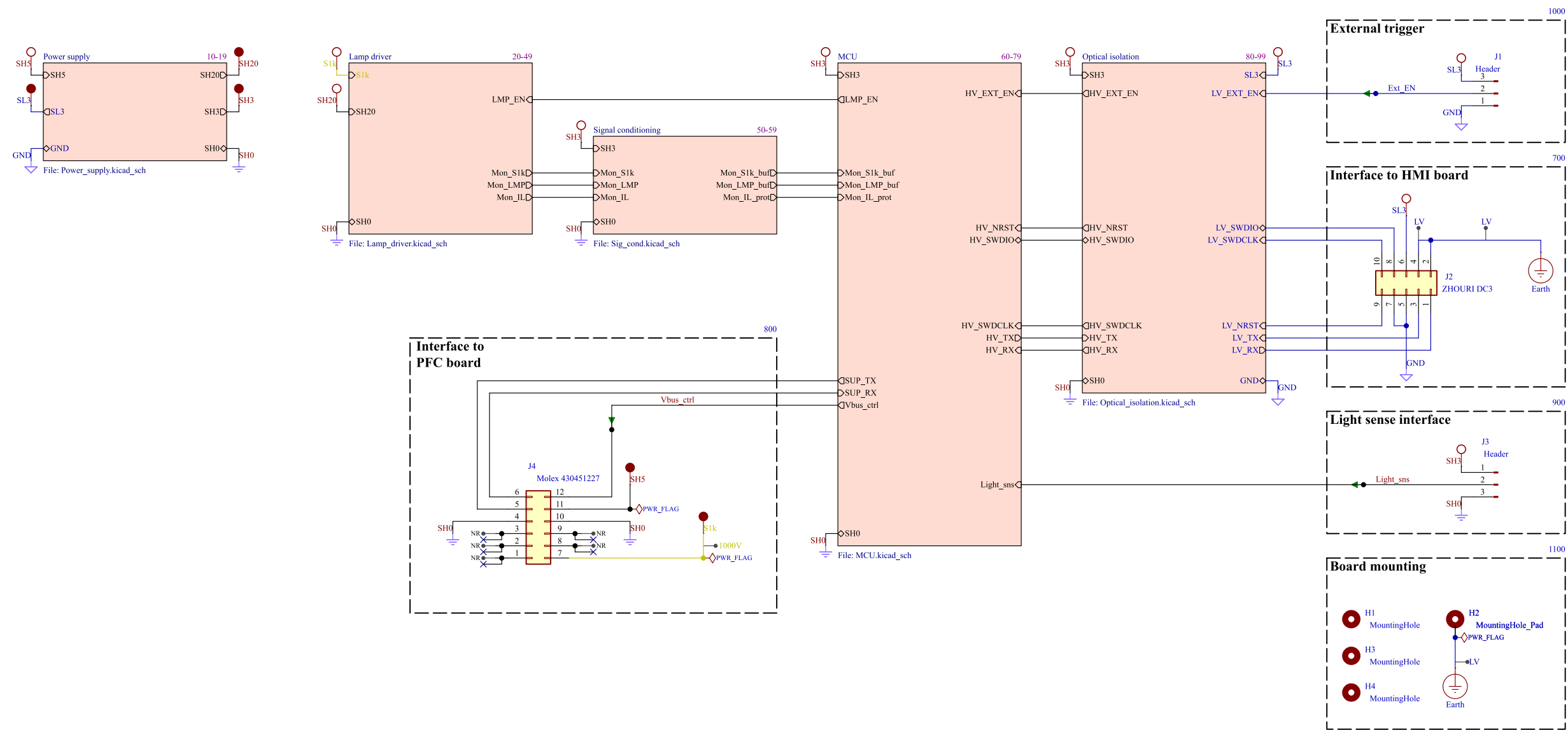
Date: 05-Sep-2026

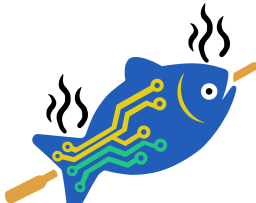
NOTE STYLING CONVENTION

DESIGN NOTE: Example text for informational design notes.	DESIGN NOTE: Example text for debug notes.	DESIGN NOTE: Example text for cautionary design notes.	DESIGN NOTE: Example text for critical design notes.	LAYOUT NOTE: Example text for critical layout guidelines.
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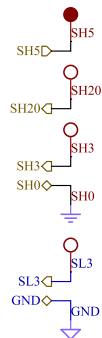
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			ChurrasCod Electronics		CHECKED		84f4bdf		
	Sheet Title:		Board Name:		Project Name:				
			Main board		1 kV strobe				
File Name:		Designer:		Date:		Revision:			
Strobe_main.kicad_sch		FR		2025-07-04		2.0.0+ (Unreleased)			
Sheet Path:		Reviewer:		Size:		Sheet:			
/				A3		1 of 8			

Block Diagram

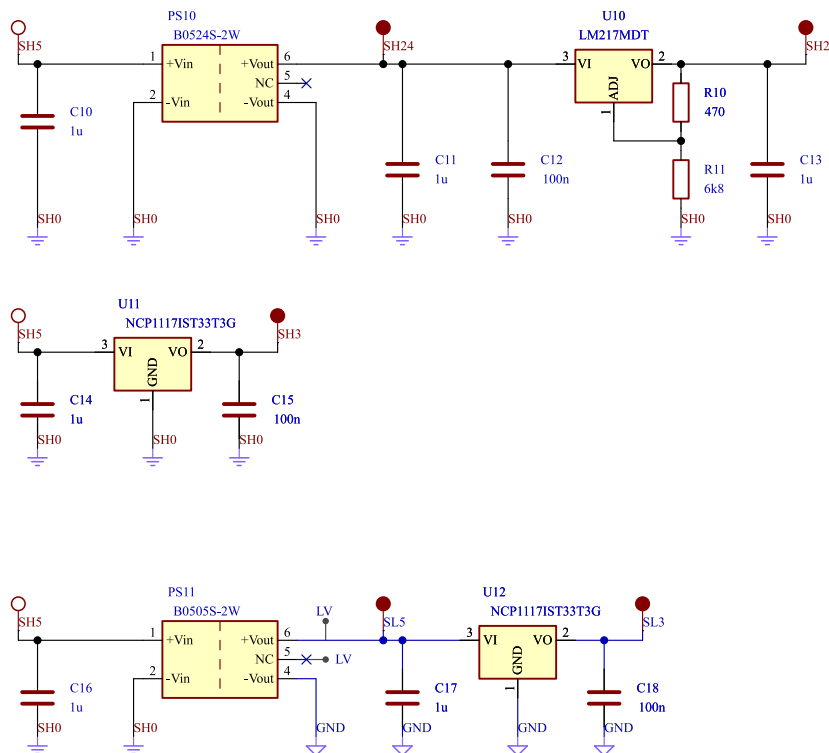


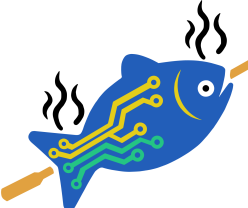
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			Board Name:			Project Name:			
			Main board			1 kV strobe			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Block Diagram		Block Diagram.kicad_sch		FR		2025-07-04		2.0.0+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
/Block Diagram/	A3					2 of 8			

BLOCK INTERFACE



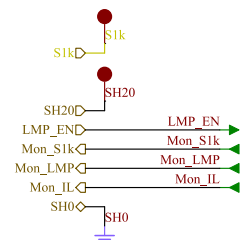
Power supply 10-19



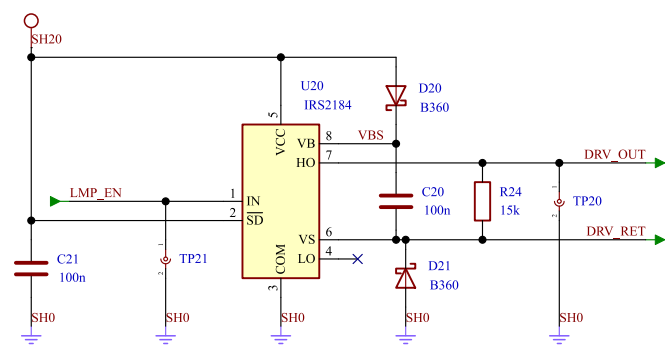
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	Sheet Path: /Block Diagram/Power supply/	File Name: Power_supply.kicad_sch	Designer: FR	Date: 2025-07-04	Revision:
			Reviewer:	Size: A4	Sheet: 3 of 8

Lamp driver

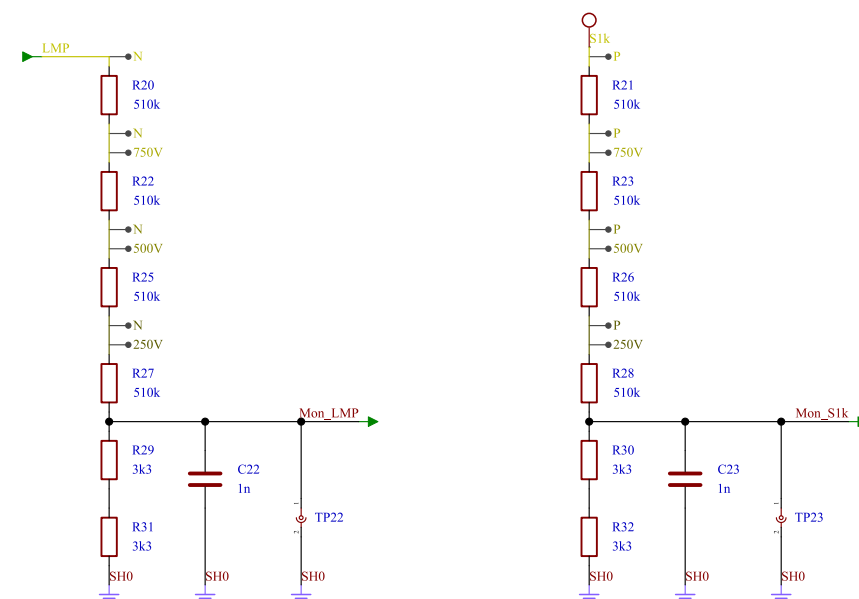
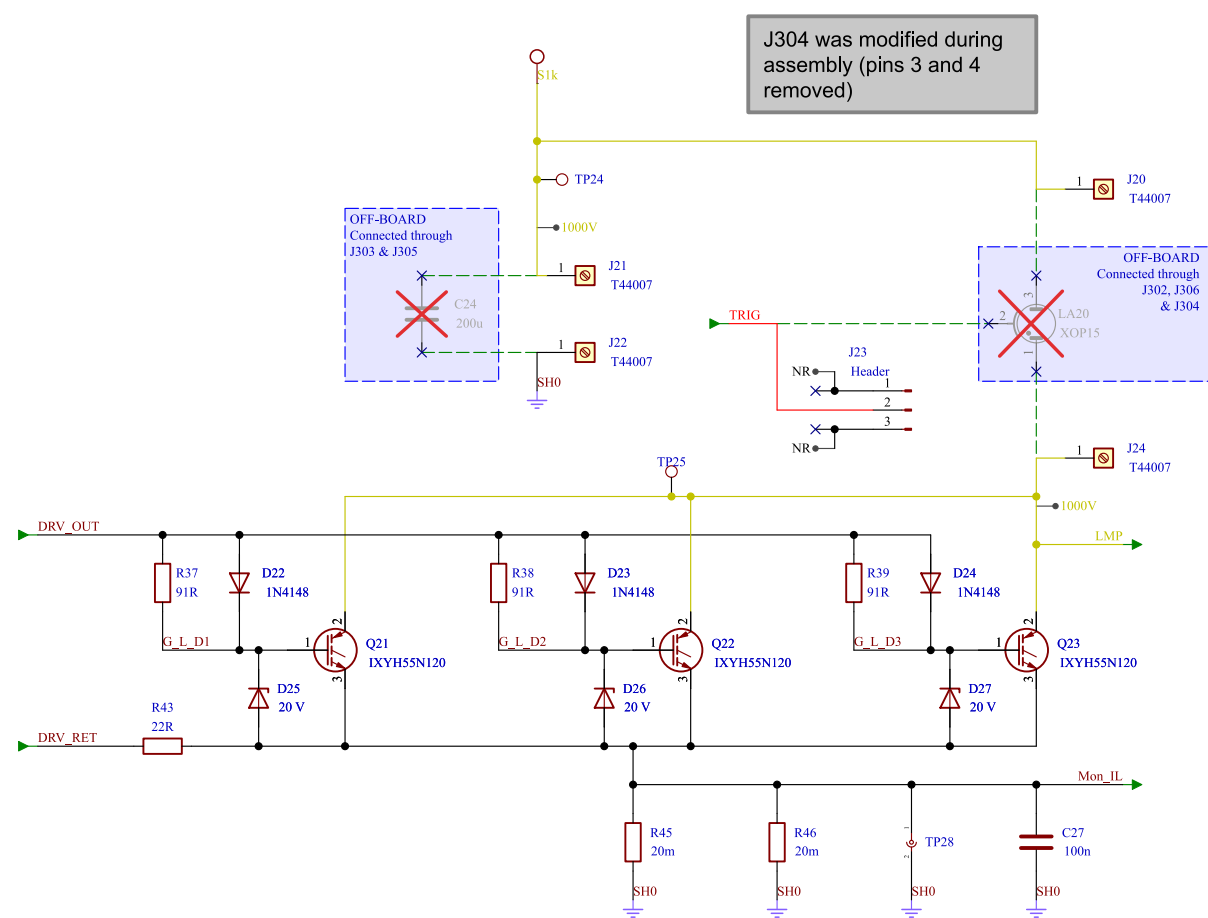
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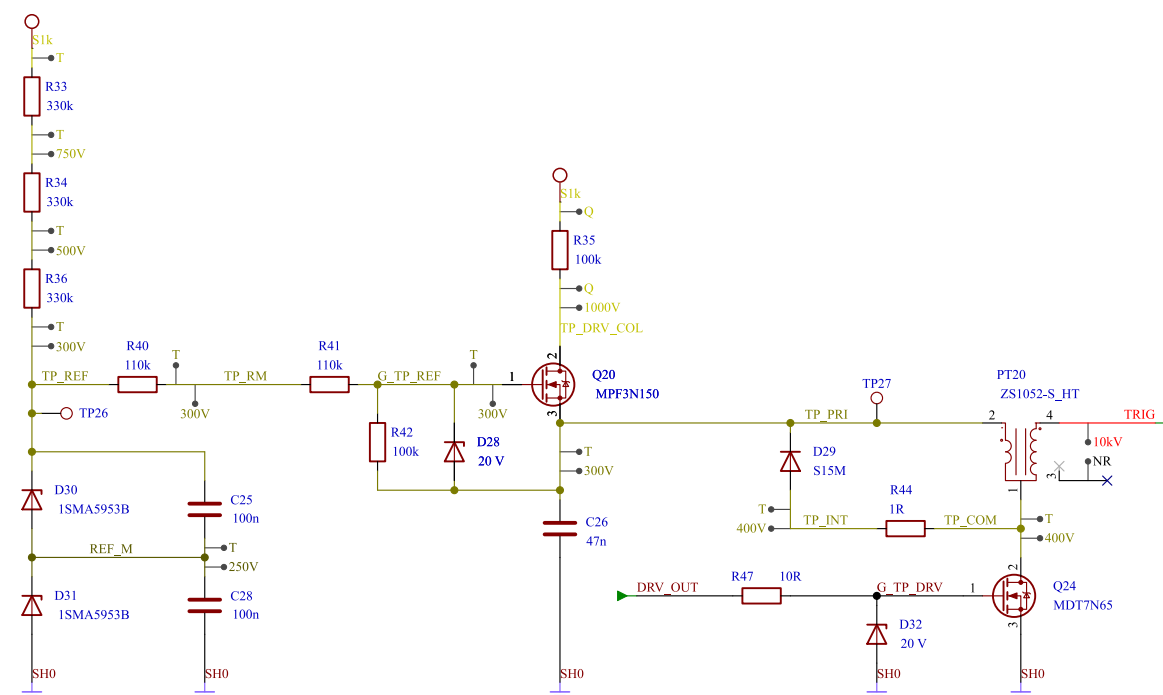
INPUT STAGE (IGBT DRIVER)



OUTPUT STAGE

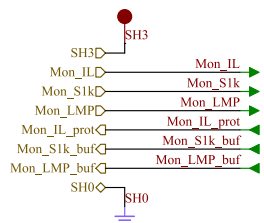


PULSE TRANSFORMER DRIVER

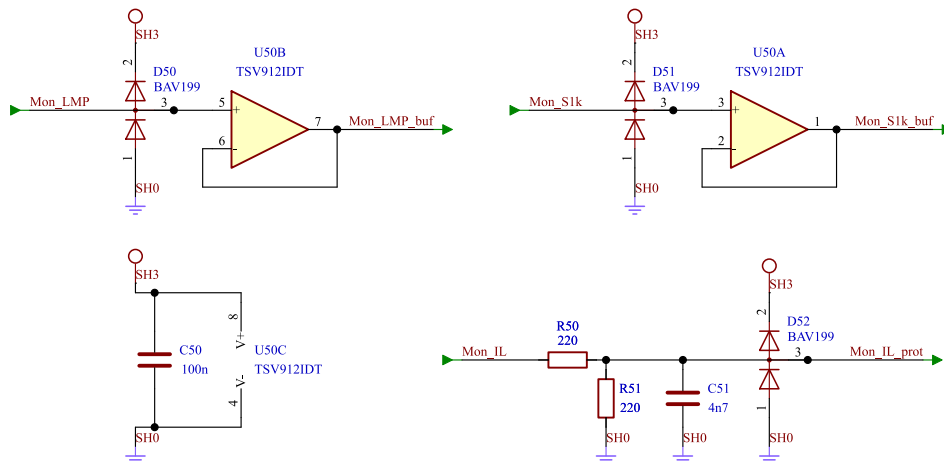


Comments:	Company:		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
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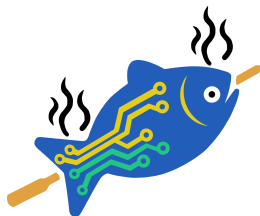
BLOCK INTERFACE



LAMP DRIVER VOLTAGES & CURRENT



Signal conditioning 50-59



ChurrasCod
Electronics

Comments:

Company:

Variant:

Git Hash:

CHECKED

84f4bdf

Board Name:

Main board

Project Name:

1 kV strobe

Sheet Title:

Signal conditioning

File Name:

Sig_cond.kicad_sch

Designer:

FR

Date:

2025-07-04

Revision:

Sheet Path:

/Block Diagram/Signal conditioning/

Reviewer:

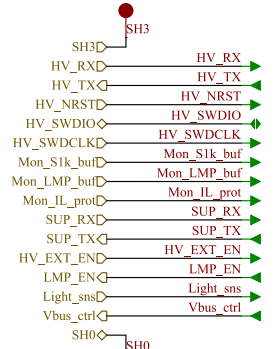
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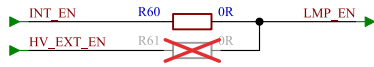
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5 of **8**

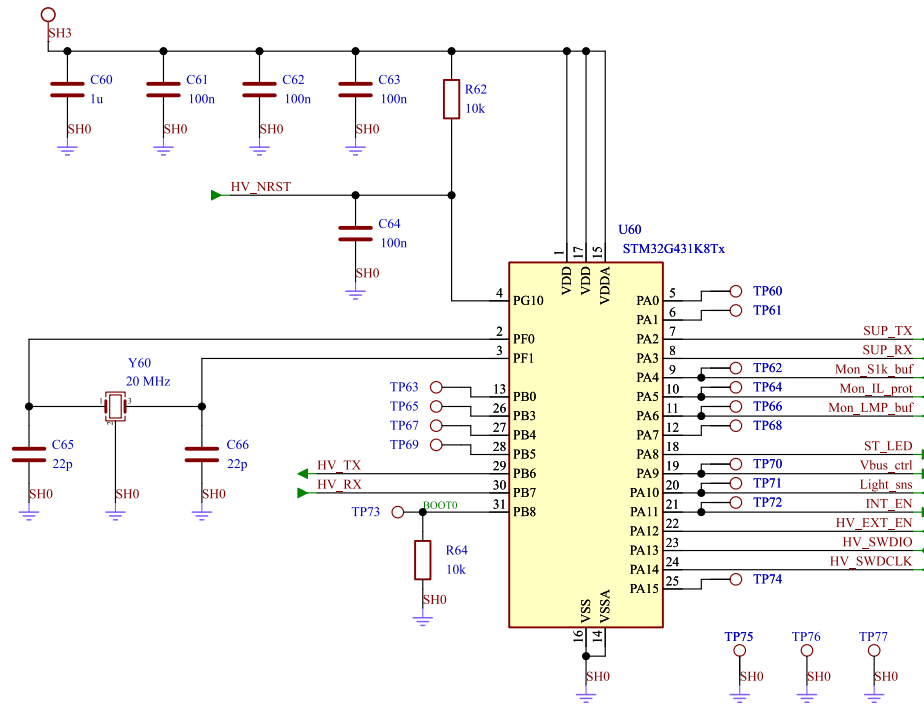
BLOCK INTERFACE



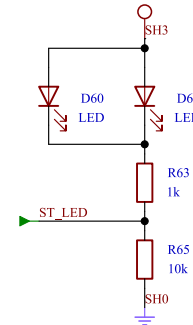
Trigger source selection



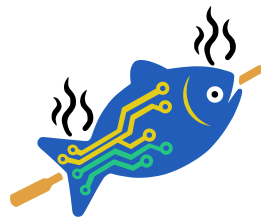
Main controller



Status LED



MCU
60-79



ChurrasCod
Electronics

Comments:

Sheet Title:

MCU

Sheet Path:

/Block Diagram/MCU/

Company:

Board Name:

Main board

File Name:

MCU.kicad_sch

Designer:

FR

Reviewer:

Variant:

CHECKED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

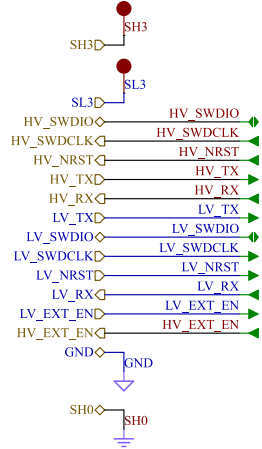
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84f4bdf

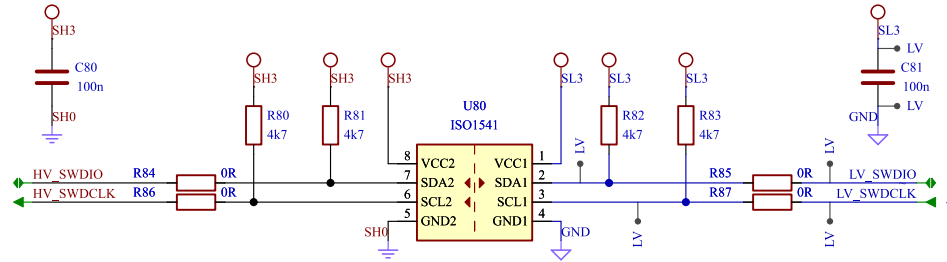
Revision:

6 of 8

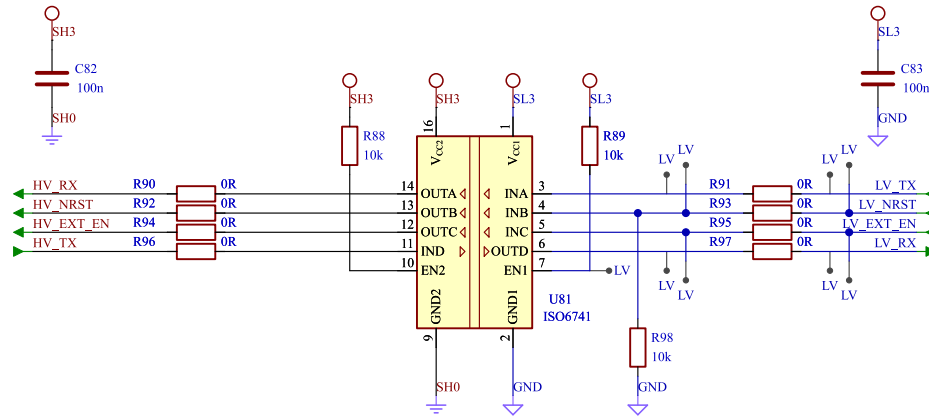
BLOCK INTERFACE



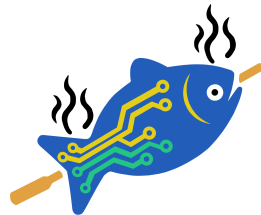
SWD optical isolator



NRST(SWD) + UART optical isolator



Optical isolation 80-99



ChurrasCod
Electronics

Comments:

Company:

Variant:

Git Hash:

CHECKED

84f4bdf

Board Name:

Main board

Project Name:

1 kV strobe

Sheet Title:

Optical isolation

File Name:

Optical_isolation.kicad_sch

Designer:

FR

Date:

2025-07-04

Revision:

Sheet Path:

/Block Diagram/Optical isolation/

Reviewer:

Size:

A4

Sheet:

7 of 8

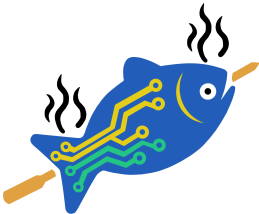
Revision History

Version 2.0.1 not found.
Version 2.0.1 not found.

- Version 2.0.0 - 2026-06-23**
- decreased impedance of lpeak measurement divider, removed buffer and fed signal directly to MCU
 - reworked gate drive resistors
 - reworked power input
 - modified some of the testpoints and added extra "GND testpoints"
 - reworked test pulse circuit to achive higher frequency
 - added mlcc to MCU rst line

Version 1.0.0 - 2026-04-11

Migrated existing project to template (KDT)



ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-07-04	Revision: 2.0.0+ (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8